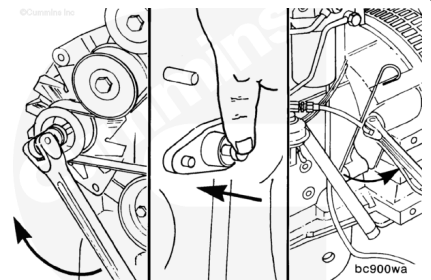


Trainings ▾

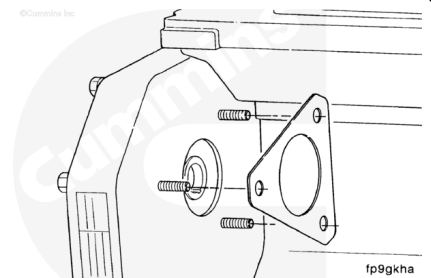
Install

Front Gear Train

Verify cylinder Number 1 is at top dead center by barring the engine slowly while pushing in on the top dead center pin.



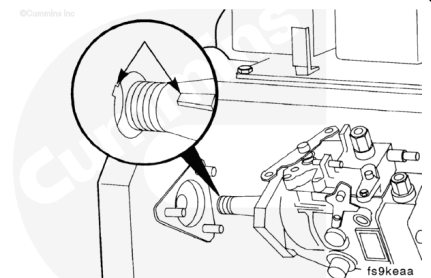
Install a new gasket.



Note : The shaft of a new or reconditioned pump is locked so the key aligns with the drive gear keyway when cylinder Number 1 is at top dead center on the compression stroke.

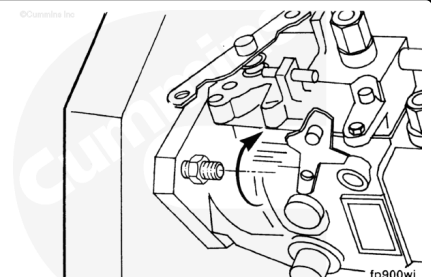
Install the pump.

Note : Fuel pumps on engines designed to meet Tier 2/Stage II Industrial emission levels do not use a timing key.



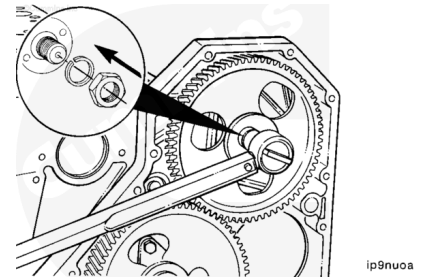
Use your hands to tighten the three mounting nuts. The pump **must** be free to move in the slots.

Note : Fuel pumps on engines designed to meet Tier 2/Stage II Industrial emissions levels have straight holes (not kidney slots).



⚠ CAUTION ⚠

Be sure the timing pin is disengaged before the final torque step to avoid damage to the timing pin.



Note : The drive shaft and drive gear bore must be clean and free from all oil before installation. Failure to make certain the drive shaft is free of oil can result in the drive gear slipping on the shaft.

Install the fuel injection pump drive gear onto the fuel pump shaft.

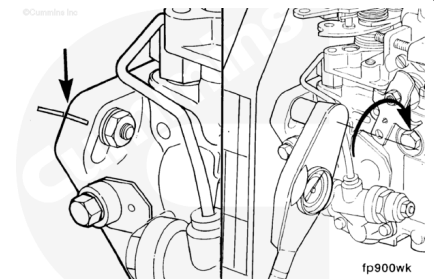
If equipped, align the timing key on the drive shaft to the slot in the fuel injection pump drive gear.

Install the pump drive shaft nut and spring washer. The pump will rotate slightly because of gear helix and clearance. This is acceptable provided the pump is free to move on the flange slots and the crankshaft does **not** move.

Torque Value: 15 to 20 n•m [11 to 15 ft-lb]

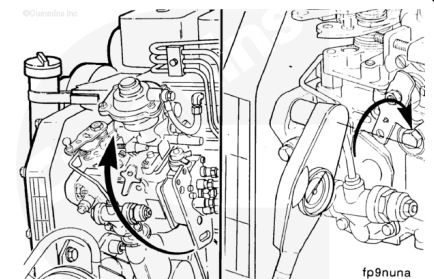
If installing the original pump, rotate the pump to align the scribe marks.

Torque Value: 24 n•m [18 ft-lb]

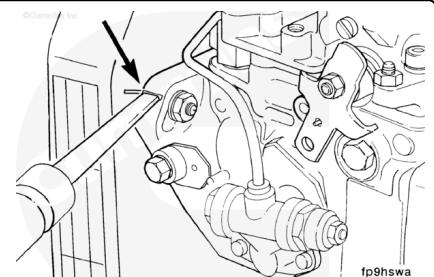


If installing a new or rebuilt pump without scribe marks, take up gear lash by rotating the pump against the direction of drive rotation. Tighten the flange mounting nuts.

Torque Value: 24 n•m [18 ft-lb]



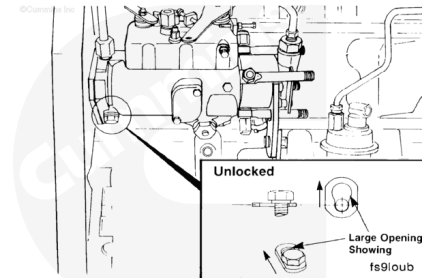
Permanently mark the injection pump flange to match the mark on the gear housing.



Lucas CAV DPA Pump and Delphi DP210

For CAV fuel injection pumps, loosen the lockscrew and position the special washer behind the lockscrew head.

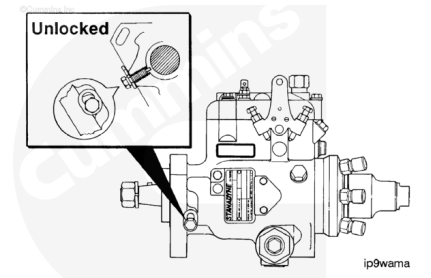
Torque Value: 20 n•m [177 in-lb]



Stanadyne DB4 Pump

For Stanadyne DB4 fuel injection pumps, loosen the lock screw and position the special washer behind the lock screw head.

Tighten the lock screw.

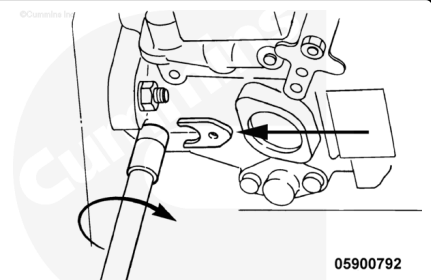


Bosch® VE

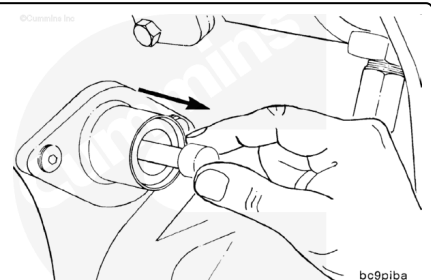
Loosen the Bosch® fuel pump lock timing screw and install the special washer that is wired to the fuel pump.

Tighten the Bosch® fuel pump lock timing screw.

Torque Value: 13 n•m [115 in-lb]



Disengage the timing pin before rotating the crankshaft.



Tighten the pump gear retaining nut.

Torque Value:

Bosch® VE (M14-1.5 nut)

1. 98 n•m [72 ft-lb]

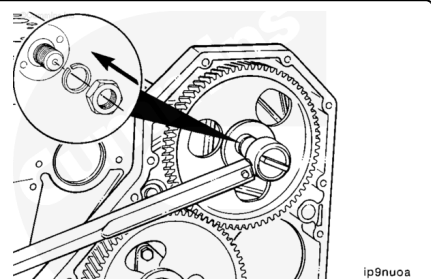
Torque Value:

Bosch® VE (M12 nut)

1. 65 n•m [48 ft-lb]

Torque Value:

Lucas CAV/DPA



1. 81 n•m [60 ft-lb]

Torque Value:

Stanadyne

1. 65 n•m [48 ft-lb]

Torque Value:

Delphi DP21

1. 93 n•m [68 ft-lb]

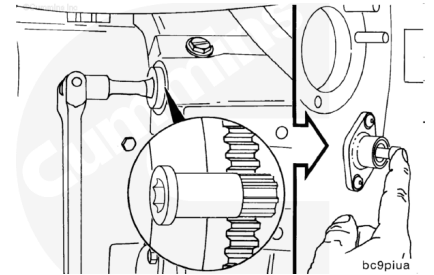
Rear Gear Train

Unlike the front gear train engine, do **not** remove the timing pin until completion of the repair. Failure to do so can result in difficult reassembly and incorrect timing of the fuel pump.

Verify cylinder number 1 is at top dead center by barring the engine slowly while pushing in on the top dead center pin.

Barring the engine is recommended from the flywheel on the rear of the engine.

Using the barring tool Part Number 3824591, rotate the crankshaft slowly while pressing on the engine timing pin to locate TDC for cylinder number 1.

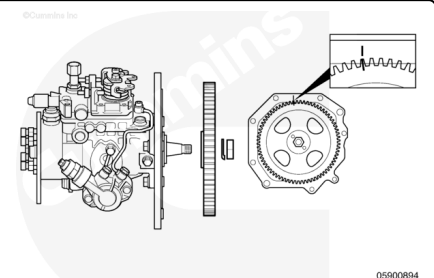


Verify the alignment mark on the fuel pump gear and the mark on the fuel pump mounting plate created during disassembly are aligned before installing the fuel pump assembly.

If the marks are **not** aligned, with the fuel injection pump locked, the fuel pump gear **must** be loosened and the marks aligned.

Tighten the pump retaining nut.

Torque Value: 98 n•m [72 ft-lb]



With a new fuel pump cover plate gasket, install the fuel pump, fuel pump mounting plate, and fuel pump gear assembly onto the rear gear housing.

Torque Value:

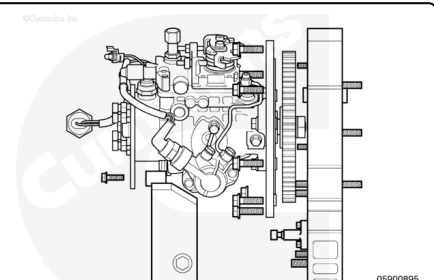
M8

1. 18 n•m [159 in-lb]

Torque Value:

M10

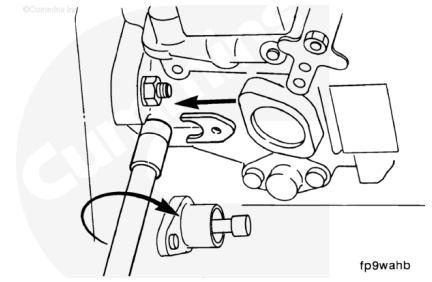
1. 30 n•m [266 in-lb]



Loosen the Bosch® fuel pump lock timing screw and install the special washer that is wired to the pump.

Tighten the Bosch® fuel pump lock timing screw.

Torque Value: 13 n•m [115 in-lb]



Last Modified: 27-Sep-2022
